

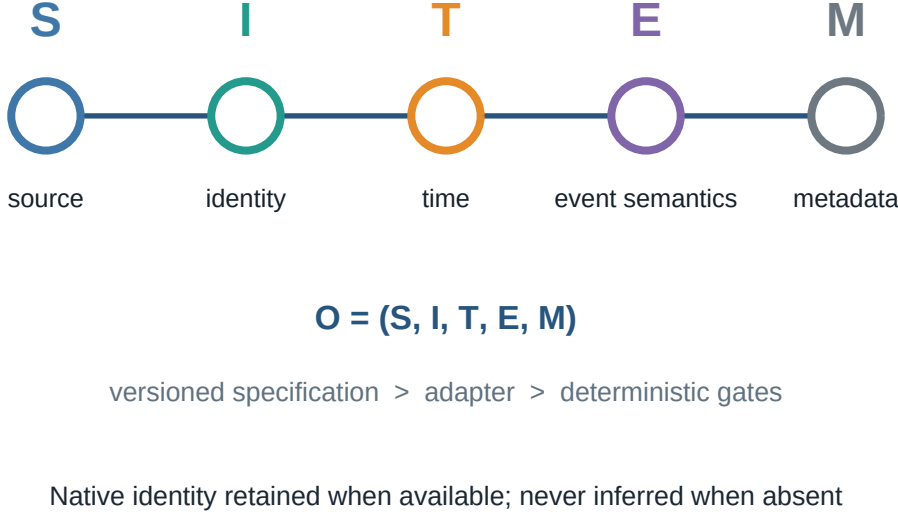
Medication exposure is a provenance-sensitive computation

A fail-closed operator preserves the difference between no documented exposure and no measurement capability

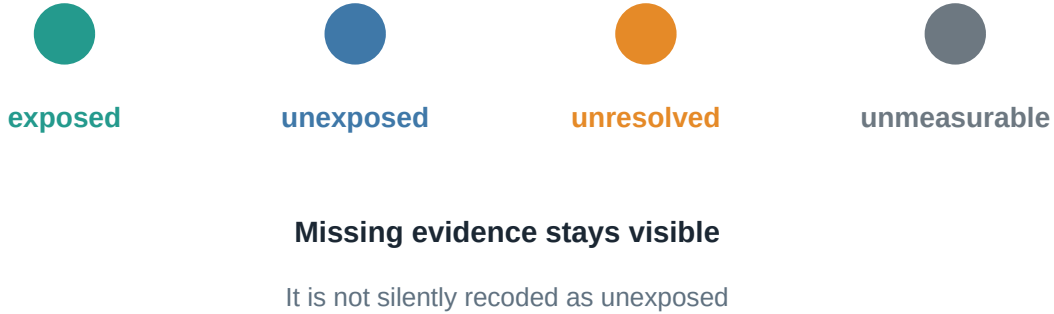
EHR MEDICATION REPRESENTATIONS

- Orders**
treatment intention
- Dispensing**
pharmacy workflow
- Administration**
documented delivery
- FHIR**
resource roles
- OMOP**
DRUG_EXPOSURE
- eICU**
non-equivalent interfaces

EXECUTABLE PROVENANCE OPERATOR



FAIL-CLOSED CLASSIFICATION



EXECUTED SEMANTIC-LOSS AUDIT

- OMOP

record existence: 37 / 37 exposed

strict administration: 37 / 37 unmeasurable
- FHIR

role / status: 6,697 positive

event semantics: 5,740 strict positive

DOWNSTREAM

